AHOPPA: Agentic Humanoid Operation through Perception, Planning, and Action

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https://hebbian-robotics.github.io/AHOPPA/ Raw data

Abstract: We study extended humanoid tasks through visual feedback and numerical motion targets, without task skills. AHOPPA connects multimodal models to the fixed SONIC controller on a simulated Unitree G1. The model inspects four drawers, pushes open ones shut and returns home while physics continues during inference. Across 64 episodes with four models at two reasoning settings, GPT-6 Astra succeeded in 12/14 manipulation attempts, taking 615 to 1492 s. Independent evaluation distinguishes physical completion from declared success and exposes falls, timeouts and false declarations. Higher effort increased physical progress for some models without consistently improving success.

Keywords: Humanoid control, multimodal agents, online evaluation

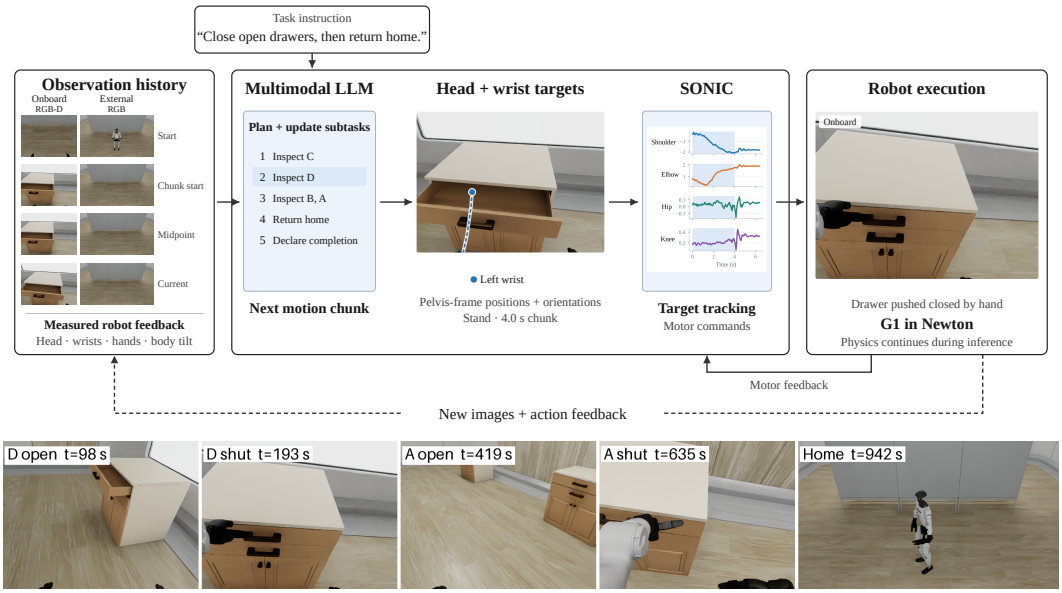


Figure 1: Agent loop and original observations from a successful Astra low episode with A and D open. Home observation: 942 s; declaration: 955 s.

1 Online humanoid operation

Room-scale humanoid operation couples navigation, inspection, contact and stopping while physics continues during inference. We test frontier models' ability to operate a humanoid through numerical references to the fixed SONIC controller [1]. Prior work spans skills [2, 3, 4], code [5, 6], visual feedback [7, 8], spatial representations [9, 10, 11] and learned action representations [12]. Our approach uses numerical commands without predefined task skills, generated policy code, value maps or keypoint constraints.

2 Interface, task and evaluation

Interface. Inputs are onboard RGB-D, external RGB, robot kinematics, depth queries, a map, the plan and feedback (Fig. 1). SONIC's gamepad commands set walking speed, direction, facing and posture; three-point tracking follows head/wrist positions and orientations. Four pose keyframes span a chosen duration; hand presets are optional. Drawer states, contacts and scores are withheld.

Task. Inspect four top drawers, push open ones shut by hand, leave closed ones undisturbed, return home and declare completion. Four passive Sektion drawers [13] occupy Simple Room [14, 15]: A/B behind lockers [16], C/D behind home; all fronts are initially hidden. Eight initializations: none (control), A, B, C, D, A+B, A+D or B+C open; all open drawers start at 0.20 m under the same instruction. Newton [17] in Isaac Lab [18] runs at 200 Hz.

Evaluation. Success requires all drawers within 2 cm of shut at $\leq$1 cm/s for 1 s, a supported standing hold within 25 cm of home and a completion declaration. Inspection, hand attribution and disturbance are scored separately. After pilots, we froze the setup and 1800 s budget. GPT-6 Astra, GPT-6.1 Sol, Claude Opus 5.5 and Gemini 4 Argon ran once per condition/effort, four concurrently.

3 Results

Model / effort	Success	Physical completion	False declarations	Falls	Mean API cost ($)	Episode time (s)	Decision latency (s)
Astra / low	6/7	6/7	0/7	1/7	9.30 / 10.57	817 / 926	9.2
Astra / high	6/7	6/7	0/7	1/7	7.65 / 8.78	998 / 1145	21.4
Sol / low	4/7	4/7	1/7	2/7	1.72 / 1.96	829 / 940	10.4
Sol / high	3/7	5/7	0/7	2/7	1.74 / 1.59	1235 / 1162	21.5
Opus / low	2/7	3/7	2/7	3/7	4.78 / 6.23	762 / 1028	13.2
Opus / high	3/7	3/7	1/7	2/7	7.22 / 8.36	1392 / 1563	36.1
Gemini / low	0/7	0/4	0/4	3/4	1.29* / –	737 / –	32.3
Gemini / high	0/7	0/5	0/5	1/5	1.59* / –	1332 / –	136.7

Table 1: Seven manipulation starts per row. Physical completion: goal reached at least once; false declarations: goal unmet when claimed. Success denominators retain API errors; other outcomes omit them. Gemini: three low-effort content-filter rejections, two high-effort request timeouts. Cost/time pairs: means over all/successful attempts. Episode time runs until termination. Costs: reconciled usage at published rates [19, 20, 21]; GPU costs excluded; * excludes unresolved usage; –: no success. Latency: median of per-attempt medians for completed decisions.

Performance. Astra led (6/7 each); 11/12 successes viewed all four drawer fronts. Sol followed with 7/14 successes, Opus with 5/14, and Gemini with none (Table 1). All 43 closures were by hand; no initially closed drawer was disturbed.

High vs. low effort. Higher effort increased decision latency roughly 2–4×; Astra's success was unchanged and Opus improved. Sol high closed more open drawers (9/10 vs. 4/10) and physically completed more tasks (5/7 vs. 4/7), but two went unreported before timeout or fall, yielding fewer successes (3/7 vs. 4/7). Another Sol high run fell before any model command.

Falls. Of eight all-closed controls, three succeeded, four fell and one had an API error. Overall, 16/19 falls lacked recorded drawer contact; other collisions remain possible. Eleven of 18 post-command falls ended in holds commanding zero translational speed with retained pose/facing targets. We hypothesize that prolonged target holds during inference contribute to instability.

Cost. Astra high matched low's success at 18% lower mean API cost. Successful runs cost less for Sol high ($1.59 vs. $1.96), but including failures, Sol low spent less per success ($3.00 vs. $4.06).

Limitations. One layout and one trial per model/effort/condition limit generalization; 41/58 completed episodes exceeded 100 ms peak physics lag. Reliability and transfer need repeated trials and hardware tests.

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